

Week 1 Lecture Note

Topic: Introduction to Google Colab and Exploring Chest X-ray Images

1. Objective

In this lesson, students will learn what Google Colab is, practice running code in Colab, understand the meaning of a dataset, load the PneumoniaMNIST dataset, and display one normal chest X-ray image and one pneumonia chest X-ray image.

2. What is Google Colab?

Google Colab is an online coding notebook that allows us to write and run Python code in a web browser. It is free to use, requires no installation, is easy to share, and is widely used for AI and data science projects.

Important parts of Google Colab:

- **Code cell:** a box where you write code
- **Run button:** the button that executes the code in a cell
- **Output area:** the place where results appear after the code is run

3. What is PneumoniaMNIST?

PneumoniaMNIST is a dataset of chest X-ray images prepared for machine learning practice. Each image is assigned a label:

- **0 = normal**
- **1 = pneumonia**

Reference:

J. Yang, R. Shi, D. Wei, Z. Liu, L. Zhao, B. Ke, H. Pfister, and B. Ni, "MedMNIST v2: A large-scale lightweight benchmark for 2D and 3D biomedical image classification," *Scientific Data*, vol. 10, no. 1, Art. no. 41, 2023.

4. First Practice in Colab

First, go to the Colab and click **New Notebook**. A new notebook will open. This is your Colab file. You can rename it by clicking the notebook title at the top of the page.

In this activity, we will download the dataset, find one image labeled as normal, find one image labeled as pneumonia, and display both images side by side.